

Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



- SafeOT-typed (ours)
- FDPI (constraint-based)
- SAC-Lagrangian (penalty-based)
- SAC + fixed penalty (penalty-based)
- typed Pareto frontier (soft ≤ 25)
- SDAC (multi-constraint SOTA)
- SRCPPO (multi-constraint SOTA)
- SAC-Pen coefficient sweep
- FDPI p_f sweep
- SAC-Lag cost-limit sweep